# Kunal Malhotra

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## SUMMARY

Engineer with 9+ years building and shipping production systems for enterprise customers in financial services and other regulated industries. Currently building AI platforms on Kubernetes: a sovereign LLM serving platform with llm-d and agentgateway, GPU sharing with HAMi, and agentic workflows that expose Kubernetes, Terraform, and SaltStack to LLM agents through custom Model Context Protocol (MCP) servers. Experienced in embedded, customer-facing delivery at IBM, McKinsey, N26, and FPT Software — running technical discovery with business and engineering stakeholders, architecting cloud landing zones and data pipelines, and deploying into live, regulated environments. Strong in Google Cloud Platform, Kubernetes, observability, and turning prototypes into reliable production systems.

## EXPERIENCE

### FPT Software

*Lead Platform Engineer* Sep 2024 – Present

- Built a sovereign LLM serving platform on Kubernetes with llm-d and agentgateway, self-hosting Hugging Face models with RAG for industry-specific use cases and keeping data and models within our own infrastructure.
- Implemented GPU sharing for AI workloads on Kubernetes with HAMi, splitting GPUs into fractional allocations with memory and compute isolation so multiple models share a single device.
- Built agentic AI workflows on custom Model Context Protocol (MCP) servers that expose Kubernetes, Terraform, and SaltStack as tools to LLM agents, letting engineers query and operate infrastructure in natural language and cutting time spent on platform tasks.
- Architected a Kubernetes-based landing zone on Google Cloud with a multi-cluster Istio service mesh, supporting 0-to-scale infrastructure patterns with Terraform IaC and full observability (multi-cluster ELK, Grafana, Thanos, Prometheus).
- Leading the design and build of a Microsoft Azure landing zone for a healthcare client's OutSystems 11 development platform, moving a 20-server, 68-database estate (1.2 TB) from a vendor-run environment into client-owned infrastructure.
- Implemented Keycloak in production on Kubernetes with the Keycloak Operator, providing OIDC/OAuth2 single sign-on for platform services and migrating existing users with their password hashes preserved.
- Led the implementation of Apache Airflow on Kubernetes to orchestrate and execute the ETL pipelines that process financial-statement data.
- Led an SRE/DevOps transformation for a multinational bank, running application assessment and implementing a DORA-aligned deployment framework across hybrid infrastructure (OpenShift, IBM Power, IBM Mainframe).
- Built a FinOps platform from scratch using OpenCost and Prometheus, achieving 20% infrastructure cost reduction across client implementations.
- Developed a VM migration solution using KubeVirt into OpenShift/AWS, enabling secure workload isolation while modernizing legacy systems.
- Interview and hire DevOps engineers for the platform team, focusing on automation via agentic AI workflows, DevOps/GitOps and cloud skills.

### McKinsey & Company

*Senior Cloud Engineer I — Cloud by McKinsey* Jan 2023 – Oct 2023

- Embedded with enterprise clients as a cloud architecture consultant, specializing in cloud-native architecture and DevOps transformation.
- Led FinOps and SRE initiatives for a multinational firm, optimizing IaaS/PaaS and OpenShift workloads and cutting cloud spend by roughly 20%.
- Designed automated testing workflows for a financial institution's MQ-based applications using Azure DevOps, improving release reliability.

**N26**
*Site Reliability Engineer* Dec 2021 – Dec 2022

- Architected and engineered the automation and infrastructure (multi-VPC architecture, site-to-site VPN) to integrate a partner bank, enabling N26 to process instant payments in a Eurozone country; the solution supports over 1 million payments per day.
- Implemented a service mesh using Istio across the heterogeneous environment, laying the foundation for a zero-trust network.
- Led a proof of concept to extend the Istio service mesh to virtual machine workloads, deploying Istio on VMs so they could join the mesh and gain end-to-end mTLS encryption, routing, and service discovery.
- Built automation for the connectivity team's AWS maintenance work, eliminating potential human error and reducing maintenance time by 95%.
- Maintained company-wide networking platform services: AWS site-to-site VPNs with external partners, client-to-site VPNs for employees, and securing of networking components (VPC, load balancers, PrivateLinks, DNS, certificate and secret management). Handled developer on-call support, incident response, and root-cause analysis.

**IBM**
*Cloud Platform Engineer* Oct 2018 – Nov 2021

- Led the migration of a multinational bank's mobile banking platform from on-prem to cloud: assessing which services could move with minimal change, re-engineering those that could not, building the landing zone, integrating with the bank's on-prem network, migrating data safely, building CI/CD pipelines, and implementing the IBM Cloud for Financial Services framework for risk and compliance.
- Led the implementation of Red Hat OpenShift at the edge for a multinational bank, enabling it to deliver services while meeting data-residency compliance requirements.
- Co-authored the “Client Reference Architecture Playbook,” an IBM Academy of Technology initiative used globally by clients to set cloud-native architecture strategy.
- Worked with the global tech sales team on a scalable KubeVirt platform strategy to modernize customer workloads across hybrid cloud environments.
- Created and delivered enablement on cloud-native technologies (Docker, OpenShift, Kubernetes, Istio, Tekton, Terraform, KubeVirt, IBM Cloud) to enterprise clients and IBM teams; delivered pilots and projects for clients across regulated industries.

*Cloud Advocate and Developer* Sep 2017 – Sep 2018

- Designed and built a proof of concept automating insurance claims via blockchain for a government client; the PoC was selected and showcased at the Arab Health Conference 2018.
- Delivered the MVP of a government project as front-end developer, establishing a solid foundation for the work.
- Built an Android application that performed facial recognition using a custom-trained IBM Watson Visual Recognition model.
- Trained 100+ developers on IBM Cloud technologies through workshops and hands-on labs.

## EDUCATION

**Birla Institute of Technology and Science, Pilani — Dubai Campus**
*Bachelor of Engineering (Honors), Computer Science* Class of 2018

- Thesis — Large-Scale Sentiment Analysis Using Apache Spark: built a framework for large-scale sentiment analysis across hundreds of nodes using Apache Spark, Apache Kafka, MLlib, Pandas, and Python; implemented the bag-of-words algorithm from scratch.

## SKILLS

**AI & GenAI:** Claude, Ollama, llm-d (distributed LLM inference on Kubernetes), agentgateway, Model Context Protocol (MCP) servers, agentic AI workflows, RAG (Milvus)

**Languages:** Python, Go, Bash

**Cloud Platforms:** Google Cloud Platform, Amazon Web Services, IBM Cloud

**Kubernetes & Orchestration:** Kubernetes (kubeadm, kubespray, k3s, kind), Red Hat OpenShift, containerd, Docker, Helm, KubeVirt, Knative, HAMi (GPU sharing), etcd operations and quorum recovery, air-gapped clusters

**Platform Engineering:** Backstage, Cluster API, Crossplane, Kyverno, External Secrets Operator

**IaC & Configuration:** Terraform, OpenTofu, Ansible, SaltStack, Consul, Vault

**Identity & API Gateways:** Keycloak (OIDC/OAuth2, Keycloak Operator, user migration with password-hash preservation), Kong.

**Service Mesh & Networking:** Istio (sidecar and ambient modes, multi-cluster, VM workloads), site-to-site VPNs, client-to-site VPNs, HAProxy, NGINX.

**CI/CD & GitOps:** Argo CD, GitHub Actions, GitLab (self-managed), Jenkins, Tekton, Azure DevOps

**Linux & Security Operations**: RHEL, Red Hat Satellite, Tenable; fleet-wide OS patching and vulnerability remediation, zero-downtime rolling node maintenance

**Observability & FinOps:** Prometheus, Grafana, Thanos, ELK Stack, Datadog, OpenCost

**Data & Storage**: Apache Airflow, Apache Kafka (KRaft), Apache Spark, PostgreSQL (CloudNativePG), Redis, MinIO, ETL pipelines, data migration

## CERTIFICATIONS

- Certified Kubernetes Application Developer (CKAD) — The Linux Foundation
- IBM Certified Application Developer – Cloud Platform V2 — IBM
- IBM Architectural Thinking — IBM

## AWARDS

- “Rookie of the Year” for technical achievements, IBM Annual Recognition event, Spain.
- Runner-Up, Junction 2018 (Europe's largest hackathon), Finland.
- 2nd Prize (Best Use of Data), Dubai Happiness Tech Sprint 2018 (Startupbootcamp & SAP).
- Developer Prize and Emirates Special Award (Best API usage), IATA NDC Hackathon, Dubai.
- Presented Packed (startup) at the IATA International Business Travel Summit 2017, Geneva.